Q.27 What is cutter radius compensation? Why is it used? Explain various codes for cutter radius compensation.

Q.28 Define automation and its role in CNC systems. Discuss emerging trends in automation and their impact on manufacturing processes.

Q.29 provide an overview of Flexible Manufacturing Systems (FMS) and Group Technology.

Q.30 Explain the concept of CAD/CAM and its significance in CNC machining. How does it streamline the design and manufacturing process?

Q.31 What are the special mechanical design features of CNC?

Q.32 Differentiate between absolute and incremental coordinate system.

Q.33 Explain the construction of LVDT.

Q.34 What is M.C.U Explain its different types.

Q.35 Write short note on canned cycles.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Identify common problems encountered in CNC machines related to mechanical, electrical, pneumatic, and electronic components. How can these problems be diagnosed and resolved using fault finding diagnosis tools?

Q.37 What are slide ways. Explain the various design features which are considered while designing a slide ways for a CNC machine.

Q.38 Explain different types of automation along with advantages and limitations.

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Roll No. ....

5th Sem / Mechatronics
Sub : CNC Machines & Automation

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 What is the function of a CNC controller?

a) Monitor machine temperature
b) Program tool paths
c) Adjust feed rates
d) Control machine operations based on programmed instructions

Q.2 In CNC machining, what does the term "tool offset" refer to?

a) The distance between the workpiece and the tool
b) The angle of the tool relative to the workpiece
c) The deviation of the tool path from the programmed path
d) The compensation applied to account for tool wear or size variations

Q.3 What type of motors are commonly used in CNC machines for precise positioning?

a) DC motors b) Stepper motors
c) Induction motors d) Hydraulic motors

Q.4 Which of the following is NOT a component of a CNC machining center?

a) Tool magazine b) Chip conveyor
c) Worktable d) Manual control panel

Q.5 What is the primary advantage of using CAD/CAM software in CNC machining?

a) Improved machine accuracy
b) Faster programming
c) Reduced machine maintenance
d) Enhanced operator safety

Q.6 What is the role of a coolant system in CNC machining?

a) Lubricating the machine components
b) Cooling the workpiece and tool
c) Monitoring machine temperature
d) Controlling chip formation

Q.7 What is the significance of the "Feed rate" in CNC machining?

a) It determines the speed of tool movement
b) It controls the spindle speed
c) It monitors coolant levels
d) It adjusts tool wear

Q.8 Which type of cutting tool is commonly used for high-speed machining in CNC operations?

a) High speed steel (HSS) tools
b) Carbide tools
c) Diamond coated tools
d) Ceramic tools

Q.9 What is the function of a tool turret in a CNC lathe?

a) Holding the workpiece
b) Changing cutting tools
c) Controlling spindle speed
d) Adjusting feed rates

Q.10 What is the primary advantages of using a CNC machining centre over a conventional machining centre?

a) Lower initial cost
b) Higher tooling flexibility
c) Lowers accuracy
d) Limited programming options

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 What does CNC stand for?

Q.12 Which component controls the movement of the cutting tool in a CNC machine?

Q.13 What is the primary advantage of using CNC machines in manufacturing?

Q.14 G-90 stands for _______.

Q.15 What is the purpose of G-Code in CNC machining?

Q.16 What type of automation is commonly used in CNC machining?

Q.17 What is the function of a tool changer in a CNC machine?

Q.18 What is the role of CAM software in CNC machining?

Q.19 Which type of motors are commonly used in CNC machines for precise positioning?

Q.20 What is the primary advantage of using CAD/CAM software in CNC machining?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Discuss the role of automatic tool changer systems in CNC machining. How do they improve efficiency and productivity?

Q.22 Explain the function of control systems in CNC machines. Compare and contrast openloop and closed loop systems.

Q.23 Describe the operation of actuators in CNC systems. How do they contribute to motion control?

Q.24 Explain the construction and application of Tachometer.

Q.25 What is a part program? Explain different part programming formula.

Q.26 What is subroutine? Why is it used?